

BIOS 755: Multilevel Models

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Introduction

- ▶ Mixed models are used for both longitudinal and multilevel data.
- ▶ Both settings involve **clustering**:
 - ▶ Longitudinal: repeated observations within subjects
 - ▶ Multilevel: units nested within higher-level units
- ▶ Multilevel data arises from study design or natural hierarchy

Multilevel Data

Examples:

- ▶ Individuals within families or communities (zip code, track tract, municipality, county, etc.)
- ▶ Students within classrooms or schools
- ▶ Patients within hospitals or practices
- ▶ Disease mapping (individuals within geographic regions)

Data may be cross-sectional or longitudinal

Multilevel Linear Models

- ▶ Also called **hierarchical linear models**
- ▶ Clustering is modeled using **random effects**
- ▶ Covariates can be measured at any level
- ▶ Example:
 - ▶ BMI: individual diet, family habits, community environment
 - ▶ ADHD: child, family, county, and state-level variables

Two-Level Setup

- ▶ Level 1 units: $i = 1, \dots, n_{1j}$
- ▶ Level 2 units: $j = 1, \dots, n_2$
- ▶ Note that j in multilevel models is akin to i in longitudinal models.
- ▶ Outcome: Y_{ij}
- ▶ Covariates: $\mathbf{X}_{ij}$

$$E(Y_{ij}) = \mathbf{X}_{ij}\beta$$

Example:

$$E(Y_{ij}) = \beta_0 + \beta_1 \text{Trt}_{ij}$$

Random Intercept Model

$$Y_{ij} = \beta_0 + \beta_1 Trt_{ij} + b_{0j} + e_{ij}$$

- ▶ $b_{0j} \sim N(0, g)$ (between-cluster variation)
- ▶ $e_{ij} \sim N(0, \sigma^2)$ (within-cluster variation)
- ▶ Observations are independent **given** b_{0j}

Important: note that b_{0j} in multilevel models is akin to b_{0i} in longitudinal models.

Sources of Variation

- ▶ Within-cluster variation: $\text{Var}(e_{ij}) = \sigma^2$
- ▶ Between-cluster variation: $\text{Var}(b_{0j}) = g$

$$\text{Corr}(Y_{ij}, Y_{kj}) = \frac{g}{g + \sigma^2}$$

- ▶ Intraclass Correlation (ICC)

General Mixed Model Form

$$Y_{ij} = \mathbf{X}_{ij}\boldsymbol{\beta} + \mathbf{Z}_{ij}\mathbf{b}_j + e_{ij}$$

- ▶ $\mathbf{b}_j$ constant within cluster j
- ▶ $E(Y_{ij}|\mathbf{b}_j) = \mathbf{X}_{ij}\boldsymbol{\beta} + \mathbf{Z}_{ij}\mathbf{b}_j$
- ▶ Conditional independence given random effects

Key Questions

- ▶ Is there between-cluster heterogeneity? ($\sigma^2 = 0$?)
- ▶ How large is the heterogeneity?
- ▶ What is the within-cluster correlation?

For the following questions, we'll consider this model:

$$Y_{ij} = \mathbf{X}_{ij}\boldsymbol{\beta} + b_{0j} + e_{ij}$$

Analysis with a Two-level Linear Model

- ▶ Is there a significant between clinic heterogeneity?
 - ▶ Can we quantify the between-clinic heterogeneity?
 - ▶ What is the degree of within-clinic correlation?
-
- ▶ What sample sizes, i.e., n_2 and n_{1j} , would be needed to answer these questions?

Analysis with a Two-level Linear Model

- ▶ Is there a significant between clinic heterogeneity?
 - ▶ Is $g = 0$? We can test this with the covtest option.
- ▶ Can we quantify the between-clinic heterogeneity?
- ▶ What is the degree of within-clinic correlation?
- ▶ What sample sizes, i.e., n_2 and n_{1j} , would be needed to answer these questions?

Analysis with a Two-level Linear Model

- ▶ Is there a significant between-clinic heterogeneity?
 - ▶ Is $g = 0$? We can test this with the covtest option.
 - ▶ Can we quantify the between-clinic heterogeneity?
 - ▶ We can give a range of intercept values with $\hat{\beta}_0 \pm 1.96 \times \sqrt{\hat{g}}$.
 - ▶ What is the degree of within-clinic correlation?
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- ▶ What sample sizes, i.e., n_2 and n_{1j} , would be needed to answer these questions?

Analysis with a Two-level Linear Model

- ▶ Is there a significant between-clinic heterogeneity?
 - ▶ Is $g = 0$? We can test this with the covtest option.
- ▶ Can we quantify the between-clinic heterogeneity?
 - ▶ We can give a range of intercept values with $\hat{\beta}_0 \pm 1.96 \times \sqrt{\hat{g}}$.
- ▶ What is the degree of within-clinic correlation?
 - ▶ Observations for a given clinic have an intraclass correlation (ICC) equal to

$$\text{Corr}(Y_{ij}, Y_{kj}) = \frac{g}{g + \sigma^2}.$$

- ▶ What sample sizes, i.e., n_2 and n_{1j} , would be needed to answer these questions?

Including Level-2 Covariates and Two Random Effects

$$Y_{ij} = \beta_0 + \beta_1 Trt_{ij} + \beta_2 Type_j + \beta_3 D_j + b_{0j} + b_{1j} Trt_{ij} + e_{ij}$$

- ▶ $Type_j$ and D_j are level-2 covariates
- ▶ D_j could represent the density of clinics in the area
- ▶ b_{0j} is a random intercept
- ▶ b_{1j} is a random slope for treatment

Same Model in Hierarchical Notation

$$Y_{ij} = \beta_{0j} + \beta_{1j} \text{Trt}_{ij} + e_{ij}$$

$$\beta_{0j} = \beta_0 + \beta_2 \text{Type}_j + \beta_3 D_j + b_{0j}$$

$$\beta_{1j} = \beta_1 + b_{1j}$$

$$\begin{pmatrix} b_{0j} \\ b_{1j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, G\right), \quad e_{ij} \sim N(0, \sigma^2)$$

- ▶ The level-2 covariates explain part of the between-cluster variation
- ▶ The two random effects allow clusters to differ in baseline outcome and in treatment effect

Three-Level Structure

- ▶ Level 1: occasions (i)
- ▶ Level 2: patients (j)
- ▶ Level 3: clinics (k)

$$Y_{ijk} = \mathbf{X}_{ijk}\boldsymbol{\beta} + \mathbf{Z}_{ijk}^{(3)}\mathbf{b}_k^{(3)} + \mathbf{Z}_{ijk}^{(2)}\mathbf{b}_{jk}^{(2)} + e_{ijk}$$

- ▶ Covariates can be measured at any level
- ▶ Random effects at levels 2 and 3

Three-Level Model

$$Y_{ijk} = \mathbf{X}_{ijk}\boldsymbol{\beta} + \mathbf{Z}_{ijk}^{(3)}\mathbf{b}_k^{(3)} + \mathbf{Z}_{ijk}^{(2)}\mathbf{b}_{jk}^{(2)} + e_{ijk}$$

Example:

$$Y_{ijk} = \beta_0 + \beta_1 t_{ijk} + \beta_2 (Trt_{jk} \times t_{ijk}) + \beta_3 X_{jk} + \beta_4 X_k + b_{0k}^{(3)} + b_{0jk}^{(2)} + e_{ijk}$$

where:

$$b_{0jk}^{(2)} \sim N(0, G^{(2)})$$

$$b_{0k}^{(3)} \sim N(0, G^{(3)})$$

$$e_{ijk} \sim N(0, \sigma^2)$$

Three-Level Model (Hierarchical Form)

Level 1 (Occasions):

$$Y_{ijk} = \beta_{0jk} + \beta_{1jk}t_{ijk} + e_{ijk}$$

Level 2 (Patients within Clinics):

$$\beta_{0jk} = \beta_{0k} + \beta_3X_{jk} + b_{0jk}^{(2)}$$

$$\beta_{1jk} = \beta_1 + \beta_2Trt_{jk}$$

Level 3 (Clinics):

$$\beta_{0k} = \beta_0 + \beta_4X_k + b_{0k}^{(3)}$$

Variance and Correlation

$$\text{Var}(Y_{ijk}) = G^{(3)} + G^{(2)} + \sigma^2$$

$$\text{Corr}(Y_{ijk}, Y_{i'jk}) = \frac{G^{(2)} + G^{(3)}}{G^{(2)} + G^{(3)} + \sigma^2}$$

$$E(Y_{ijk}) = \mathbf{X}_{ijk}\boldsymbol{\beta}$$

- ▶ Software:
 - ▶ SAS: PROC MIXED
 - ▶ R: lmer

Nesting

- ▶ Nesting random effects gets very challenging computationally, even for linear mixed models.
- ▶ To help with the computations, use

```
random school;  
random class(school);  
repeated / subject=person(class*school) type=cs;
```
- ▶ Instead of

```
random school;  
random class(school);  
random person(class*school);
```